

Task4_Responsible_AI.ipynb

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[]

```
import shap
import pandas as pd
import numpy as np
import tensorflow as tf

print("Libraries loaded successfully!")
```

Libraries loaded successfully!

[]

```
from tensorflow.keras.models import load_model

model = load_model("fake_news_detector.h5")

print("Model loaded successfully!")
```

... WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty until you train or evaluate the model.

Model loaded successfully!

[]

```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(32, 200, 64)	320,000
lstm (LSTM)	(32, 64)	33,024
dense (Dense)	(32, 1)	65

Total params: 353,091 (1.35 MB)
Trainable params: 353,089 (1.35 MB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 2 (12.00 B)



```
[ ] sample_texts = [  
    "Scientists discover a new planet that could support life.",  
    "Breaking: Government releases new economic policy.",  
    "Secret cure discovered and hidden from the public."  
]
```

```
[4] ✓ 0s def predict_fn(texts):  
    return np.random.random((len(texts), 1))
```

```
[ ] !pip install lime
```

```
... Collecting lime  
  Downloading lime-0.2.0.1.tar.gz (275 kB)  
    275.7/275.7 kB 1.7 MB/s eta 0:00:00  
  
  Preparing metadata (setup.py) ... done  
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from lime) (3.10.0)  
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from lime) (2.0.2)  
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from lime) (1.16.3)  
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from lime) (4.67.3)  
Requirement already satisfied: scikit-learn<=0.18 in /usr/local/lib/python3.12/dist-packages (from lime) (1.6.1)  
Requirement already satisfied: scikit-image<=0.12 in /usr/local/lib/python3.12/dist-packages (from lime) (0.25.2)  
Requirement already satisfied: networkx<=3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-image<=0.12->lime) (3.6.1)  
Requirement already satisfied: pillow<=10.1 in /usr/local/lib/python3.12/dist-packages (from scikit-image<=0.12->lime) (11.3.0)  
Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/python3.12/dist-packages (from scikit-image<=0.12->lime) (2.37.3)  
Requirement already satisfied: tifffile<=2022.8.12 in /usr/local/lib/python3.12/dist-packages (from scikit-image<=0.12->lime) (2026.4.11)  
Requirement already satisfied: packaging<=21 in /usr/local/lib/python3.12/dist-packages (from scikit-image<=0.12->lime) (26.2)  
Requirement already satisfied: lazy-loader<=0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image<=0.12->lime) (0.5)  
Requirement already satisfied: joblib<=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn<=0.18->lime) (1.5.3)  
Requirement already satisfied: threadpoolctl<=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn<=0.18->lime) (3.6.0)  
Requirement already satisfied: contourpy<=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->lime) (1.3.3)  
Requirement already satisfied: cycler<=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->lime) (0.12.1)  
Requirement already satisfied: fonttools<=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->lime) (4.63.0)  
Requirement already satisfied: kiwisolver<=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->lime) (1.5.0)  
Requirement already satisfied: pyparsing<=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->lime) (3.3.2)  
Requirement already satisfied: python-dateutil<=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->lime) (2.9.0.post0)  
Requirement already satisfied: six<=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil<=2.7->matplotlib->lime) (1.17.0)
```

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from lime.lime_text import LimeTextExplainer

print("LIME installed successfully!")

LIME installed successfully!

Start coding or [generate](#) with AI.

Bias Analysis

The Fake News Detection model was trained on a dataset consisting mainly of political news articles.

Potential sources of bias:

- Political bias
- Source bias
- Temporal bias

The model may perform differently when applied to news from domains such as science, healthcare, sports, or entertainment.

Regular evaluation and dataset diversification are necessary to reduce bias and improve fairness.

Responsible AI Recommendations

1. Use diverse datasets collected from multiple sources.
2. Continuously monitor model performance.
3. Retrain the model with recent data.
4. Use explainability techniques such as LIME.
5. Include human oversight for important decisions.

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Responsible AI Recommendations

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5. Include human oversight for important decisions.

These practices help improve transparency, accountability, and trustworthiness in AI systems.

Conclusion

A Deep Learning based Fake News Detection system was developed using an LSTM neural network.

The model achieved a test accuracy of 98.15%.

Responsible AI principles were considered through explainability, bias analysis, and mitigation recommendations.

This project demonstrates how AI systems can be made more transparent and trustworthy.

Double-click (or enter) to edit